

LUAD Survival Analysis Project

Understanding SHAP and the Multi-Modal Fusion Model

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A plain-English deep-dive into model interpretability and multi-stream fusion

What this document covers: Two things that are critical to understand before building the final models. First, what SHAP does and why it matters clinically. Second, a deep, slow, plain-English explanation of the multi-modal fusion model — what it is, why we build it the way we do, and how it produces interpretable predictions for individual patients.

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Part 1 — What is SHAP and what does it do?

The problem with neural networks

Our DeepSurv and fusion models output one number — a risk score. But they don't tell you *why*. It's like a doctor saying 'you have high risk' without explaining what led to that conclusion. Clinicians can't trust or use a black box. SHAP solves this.

What SHAP actually does — the simple version

SHAP answers one question for every feature and every patient:

"How much did THIS specific feature push the risk score up or down for THIS specific patient?"

A concrete example — Patient A:

```
Risk score = 7.2 (HIGH RISK)

SHAP breakdown:

EGFR expression: +1.8 <- pushed risk UP strongly
TP53 expression: +1.2 <- pushed risk UP
CD8+ T cells (immune): -0.9 <- pushed risk DOWN (protective)
DKK1 dysregulation: +0.7 <- pushed risk UP
Age: +0.4 <- pushed risk UP slightly
...

Sum of all SHAP values = 7.2 <- adds up to the final risk score
```

Every feature gets a number. Positive = it increased the risk. Negative = it decreased the risk. The magnitude tells you how much. Crucially, all SHAP values sum to exactly the final risk score — the explanation is complete and mathematically consistent.

What SHAP gives us at the population level

When you run SHAP on all 478 patients you get a beeswarm plot — the standard figure in every survival analysis paper. It shows:

```
Top 30 features ranked by average importance across all patients:

EGFR ||| average |SHAP| = 2.1 <- most important
CD8+ T cell ||| average |SHAP| = 1.8
```

DKK1 ||||| average |SHAP| = 1.4

...

Each dot = one patient

Red dots = high feature value

Blue dots = low feature value

Position on x-axis = SHAP value (how much it affected risk)

This tells you globally which features matter most across the whole cohort. It answers: 'Which genes, immune cells, and dysregulation patterns are the main drivers of survival in LUAD patients?'

Part 2 — Can we explain the output for a specific patient?

Yes — and this is the most clinically exciting part. For a single patient, after the fusion model outputs their risk score, we can produce a full explanation showing exactly which biological signals drove the prediction.

Example patient report:

Patient: TCGA-05-4244

Risk Score: 8.3 (HIGH RISK — top 15% of cohort)

Why this patient is high risk:

IMMUNE SYSTEM (dominant signal — contributing most to risk):

x CD8+ T cells very low -> +1.8 risk (immune cannot fight tumour)

x Tregs high -> +0.9 risk (immune suppression active)

v NK cells moderate -> -0.6 risk (some natural killer activity)

DYSREGULATION:

x EGFR highly dysregulated -> +1.4 risk (growth pathway overactive)

x TP53 silenced -> +1.1 risk (tumour suppressor lost)

v GSTT2 preserved -> -0.3 risk (detox pathway intact)

EXPRESSION:

x HOXB9 overexpressed -> +0.9 risk (known LUAD oncogene)

x DKK1 overexpressed -> +0.7 risk (Wnt pathway disrupted)

DOMINANT SIGNAL: Immune suppression + high dysregulation

BIOLOGICAL STORY: Tumour has drifted far from normal tissue AND successfully suppressed immune detection.

This is the worst-case combination.

Can we say the immune system is weak?

Yes — exactly. If the SHAP values show CD8+ T cells contributed +1.8 to risk (high risk because they are LOW) and Tregs contributed +0.9 to risk (high risk because they are HIGH — suppressing immune), then you can directly say:

"For this patient, the dominant driver of high risk is immune suppression — the tumour has successfully evaded immune detection."

And for a different patient where expression dominates:

"For this patient, the dominant driver is molecular dysregulation — the tumour has drifted far from healthy lung biology."

The clinical impact

Without SHAP	With SHAP
Risk score = 8.3	High risk driven by immune suppression and EGFR dysregulation
Useless to a doctor	Actionable — suggests immunotherapy to restore CD8+ T cells
Black box	Biological story a clinician can understand and act on

Part 3 — Why do we have three separate data streams?

Think about predicting whether someone will survive cancer. You could look at this from three completely different angles — and these angles measure different things. They are not redundant.

Angle 1 — What genes are active in the tumour right now?

Some genes are more active, some less active. This is the RNA-seq expression data we already have. It tells us: what is the molecular activity profile of this tumour today?

Angle 2 — How different is the tumour from a healthy lung cell?

A healthy lung cell has a normal pattern of gene activity. A cancer cell has a disrupted pattern. How disrupted? That is the dysregulation score — Sonica computes this by comparing each patient's tumour to healthy lung tissue from GTEx. It tells us: how far has this tumour drifted from what it should be?

Angle 3 — What is the immune system doing?

Are there lots of cancer-fighting immune cells in the tumour? Or has the tumour suppressed them? That is the immune features from CIBERSORT. It tells us: can the immune system see and fight this tumour?

A patient could have:

Normal gene expression BUT high dysregulation AND low immune response

High expression changes AND normal dysregulation AND strong immune response

Every combination tells a different biological story. That is why we keep the three streams separate and combine them smartly — not just dump them together.

Part 4 — Why run individual Cox models before the fusion?

This is a common point of confusion. We run individual Cox models on each stream as **baselines** — not as part of the fusion. They are completely separate.

The chef analogy:

Before building a combined recipe, a good chef first tastes each ingredient separately. Does this ingredient have any flavour on its own? Is it worth including?

Model 3 – Cox on dysregulation alone

Question: does dysregulation predict survival by itself?

If C-index = 0.50 -> no signal -> maybe reconsider including it

If C-index = 0.65 -> real signal -> definitely include it

Model 4 – Cox on immune features alone

Question: does immune landscape predict survival by itself?

Same logic applies

These are diagnostic models — they tell us whether each ingredient is worth using. They are NOT the fusion model. They are separate, simple, quick baselines that sit in the model comparison table and justify the design of the fusion model. The fusion model is built AFTER these, combining all streams together.

The complete model ladder:

Model	Features	Purpose
Cox Clinical (Model 1)	age, stage, gender	Clinical standard of care benchmark
Cox-Lasso Expression (Model 2)	1,000 genes	Does expression alone help?
Cox-Lasso Dysregulation (Model 3)	Dysregulation z-scores	Does drift from normal help?
Cox-Lasso Immune (Model 4)	CIBERSORT + ssGSEA	Does immune landscape help?
DeepSurv Expression (Model 5)	1,000 genes	Non-linear expression model
Multi-modal Fusion (Model 6)	ALL streams combined	The paper's core novelty

Part 5 — The fusion model, explained slowly from scratch

The analogy: three doctors giving their opinion

Imagine a patient walks in and three specialist doctors examine them.

Doctor 1 — Gene Expression Specialist

Looks at a report with 1,000 numbers — the gene activity levels. After studying it, she says: 'Based on the expression pattern, I give this patient a risk score of 7 out of 10.'

Doctor 2 — Dysregulation Specialist

Looks at how far the tumour has drifted from normal. He says: 'This tumour has drifted very far from healthy lung tissue. Risk score: 8 out of 10.'

Doctor 3 — Immune Specialist

Looks at the immune cell composition. She says: 'The immune system is completely suppressed here. Risk score: 9 out of 10.'

Now a fourth person — the **decision maker** — listens to all three and has to give a final risk score. But the decision maker is smart. They know that for some patients, the immune picture matters most. For others, the dysregulation matters most. So they do not just average the three scores — they **weight** them based on which specialist they trust most for this particular patient.

That is the fusion model. Three encoders (the specialists) plus one attention layer (the decision maker).

Translating to code — the encoders:

Each 'doctor' is a small neural network called an **encoder**. It takes raw input and compresses it into a compact summary.

Doctor 1 = Expression Encoder

Input: 1,000 gene expression values (1,000 numbers)

Output: 32 numbers (a compressed summary of expression signal)

Doctor 2 = Dysregulation Encoder

Input: dysregulation z-scores for the same genes

Output: 32 numbers (a compressed summary of how far tumour drifted)

Doctor 3 = Immune Encoder

Input: 22 immune cell fractions + ssGSEA immune scores

Output: 32 numbers (a compressed summary of immune landscape)

Each encoder takes its raw input (hundreds of numbers) and compresses it into 32 numbers. Those 32 numbers are the encoder's 'summary opinion' about survival risk based on its data stream. The 32 is a bottleneck — the encoder is forced to compress only the most important information. This is what the neural network learns during training.

Part 6 — The attention mechanism — the decision maker

After the three encoders produce their summaries, the attention mechanism decides how much to trust each one for each patient. This is the key innovation.

How attention weights work:

Patient A – highly dysregulated tumour, low immune infiltration:

Expression summary: [0.2, 0.8, 0.1, ...] 32 numbers

Dysregulation summary: [0.9, 0.3, 0.7, ...] 32 numbers

Immune summary: [0.1, 0.1, 0.2, ...] 32 numbers

Attention learns these weights for Patient A:

Weight for expression: 0.2 (20% trust)

Weight for dysregulation: 0.6 (60% trust)

Weight for immune: 0.2 (20% trust)

Final summary = $0.2 \times \text{expression} + 0.6 \times \text{dysregulation} + 0.2 \times \text{immune}$

Patient B – normal dysregulation, strong immune activity:

Attention learns DIFFERENT weights for Patient B:

Weight for expression: 0.1 (10% trust)

Weight for dysregulation: 0.2 (20% trust)

Weight for immune: 0.7 (70% trust) <- immune dominates here

The weights are different for every patient and are learned automatically during training. The model figures out: 'For patients like A, the dysregulation signal is most informative.' For another patient the immune signal might dominate completely. This is the power of attention — it is not a fixed formula. It adapts to each patient.

Why attention here but not on the 1,000 genes?

Earlier in this project we decided not to add attention directly over the 1,000 genes. The reason is simple: attention over 1,000 genes needs to learn 1,000 weights per patient — we only have 478 patients. There is not enough data to learn those weights reliably.

But attention over 3 streams only needs to learn 3 weights per patient. That is completely manageable with 478 patients, and it is biologically motivated — we genuinely believe different streams matter differently for different patients. This is the right place for attention.

Part 7 — The complete architecture

FUSION MODEL — COMPLETE PICTURE

=====

Expression (1,000 genes) -> MLP Encoder 1 -> 32 numbers -+

|

Dysregulation (z-scores) -> MLP Encoder 2 -> 32 numbers -> Attention

| layer

Immune (fractions+ssGSEA) -> MLP Encoder 3 -> 32 numbers -+

|

Weighted combo (32 numbers)

|

Final layer -> 1 risk score

|

Cox partial likelihood loss

|

Train: learn to rank patients correctly

Each encoder's internal structure:

Each MLP encoder is a small neural network:

Input (e.g. 1,000 genes)

|

Linear layer -> 256 neurons

BatchNorm -> stabilise training

ReLU -> non-linearity (negative -> 0)

Dropout 0.3 -> regularisation (prevent overfitting)

|

Linear layer -> 64 neurons

BatchNorm

ReLU

Dropout 0.3

|

Linear layer -> 32 neurons (the compressed summary)

The attention layer's internal structure:

Three summaries arrive: [32 numbers] x 3 streams

|

Stack them: 3 x 32 matrix

|

Small network learns 3 scalar weights (one per stream)

Apply softmax: weights sum to 1.0

|

Weighted sum: $w_1 \times \text{summary}_1 + w_2 \times \text{summary}_2 + w_3 \times \text{summary}_3$

|

Output: 32 numbers (the fused patient representation)

Part 8 — Why this should beat the individual models

Each stream captures something the others miss entirely:

Stream	What it captures	What it misses
Expression alone	Absolute gene activity levels	How far tumour has drifted from normal
Dysregulation alone	Deviation from healthy tissue	Absolute activity levels of genes
Immune alone	Immune cell composition	The molecular biology of the tumour itself
Fusion (all three)	ALL of the above	Nothing — all signals combined

The biological hypothesis the fusion model tests:

When combined, the model can learn interactions like:

"A patient with HIGH dysregulation AND LOW immune infiltration has the worst survival — because the tumour has drifted far from normal AND is successfully hiding from the immune system."

Neither stream alone captures this interaction. The fusion model does. This is the central biological bet of the entire project — and it is what makes this paper novel.

Why this is clinically meaningful:

The black box problem in oncology is real. Clinicians will not act on a number they cannot explain. The fusion model + SHAP combination solves this:

- For patients where **dysregulation dominates**: the tumour has drifted far from normal. This patient might benefit from targeted therapies aimed at the specific pathways that are disrupted.
- For patients where **immune features dominate**: the immune system is suppressed. This patient is a candidate for immunotherapy to restore CD8+ T cell activity.
- For patients where **expression dominates**: specific oncogenes are overactive. This might inform targeted gene therapy or kinase inhibitor choices.

Part 9 — What SHAP shows after the fusion model is trained

Once the fusion model is trained, SHAP gives us two levels of explanation:

Level 1 — Which STREAM matters most (population level):

Average contribution per stream across all 478 patients:

Dysregulation stream: average $|SHAP|$ = 2.4 <- dominant stream

Expression stream: average $|SHAP|$ = 1.8

Immune stream: average $|SHAP|$ = 1.6

Conclusion: How far the tumour has drifted from normal tissue is the strongest overall predictor of survival.

Level 2 — Which FEATURES within each stream matter most:

Top features from dysregulation stream:

EGFR dysregulation -> average $|SHAP|$ = 0.9 (growth pathway)

TP53 dysregulation -> average $|SHAP|$ = 0.8 (tumour suppressor)

MYC dysregulation -> average $|SHAP|$ = 0.6 (proliferation)

Top features from immune stream:

CD8+ T cells -> average $|SHAP|$ = 0.7 (cancer killing)

M2 macrophages -> average $|SHAP|$ = 0.6 (immunosuppression)

Regulatory T cells -> average $|SHAP|$ = 0.5 (immune evasion)

Top features from expression stream:

H0XB9 -> average $|SHAP|$ = 0.5 (known LUAD oncogene)

DKK1 -> average $|SHAP|$ = 0.4 (Wnt pathway)

This is what Sonica interprets biologically. She takes the top SHAP features and maps them to known cancer pathways using clusterProfiler and MSigDB. She answers: 'Does the dysregulation of EGFR and the suppression of CD8+ T cells together tell a coherent biological story about immune evasion?' If yes — that becomes the paper's biological narrative.

In the Streamlit web app:

When we build the web app, a clinician inputs a patient's data and gets:

- A **risk score** with percentile rank in the cohort
- A **risk group** — low, medium, or high risk
- A **bar chart** showing which streams and features drove the prediction
- A **plain English explanation** — 'Primary risk driver: immune suppression (CD8+ T cells low, Tregs high)'
- A **survival curve** showing predicted survival probability over time

Complete Loop: Raw Data → Trained Model → Clinical Insight

Raw TCGA data -> Three encoders -> Attention fusion -> Risk score

Risk score -> SHAP analysis -> Feature contributions per patient

Feature contributions -> Biological interpretation -> Clinical decision support

This is not just a better number. It is a biological story, backed by interpretable machine learning, validated in independent cohorts.